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APPLIED RESEARCH

Adaptive Algorithm for Selecting the Optimal Trading Strategy Based on Reinforcement Learning for Managing a Hedge Fund

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ABSTRACT In hedge fund management, the ability to dynamically select optimal trading strategies is paramount for maximizing returns and mitigating risk. This paper presents a pioneering approach that integrates Reinforcement Learning (RL), specifically the Proximal Policy Optimization (PPO) algorithm, into the strategy selection process for hedge fund management. Our model considers a diverse array of strategies, including Mean Reversion and Momentum, and employs advanced mathematical frameworks to evaluate and select the strategies. By leveraging RL, our algorithm learns to adaptively adjusts strategy allocations to maximize cumulative returns while adhering to the risk constraints. We demonstrate the effectiveness of our approach through extensive backtesting and validation of historical market data, demonstrating superior performance compared to traditional methods. Nevertheless, it is important to understand that training trading agents requires a considerable amount of time, computing power, and other resources. Our research offers a novel perspective on leveraging RL to optimize strategy selection in hedge fund management and underscores the potential of AI-driven approaches in finance.

INDEX TERMS Algorithmic trading, artificial intelligence, deep reinforcement learning, hedge fund management, optimal trading strategy, proximal policy optimization (PPO) algorithm, trading policy.

I. INTRODUCTION

Hedge fund management requires the dynamic selection of trading strategies to navigate ever-changing market conditions. Traditional methods often fall short of effectively adapting to market dynamics. In this study, we introduce a Reinforcement Learning-based approach, specifically utilizing the Proximal Policy Optimization algorithm, to optimize strategy selection in hedge fund management.

Forecasting the behavior of financial markets poses a number of challenges, including the degradation of predictive ability on hold-out samples and real trading due to non-stationary and noisy data structures. The scientific community and practitioners are actively testing and integrating machine learning and reinforcement learning (RL) algorithms to solve specific problems. It has been demonstrated that such algorithms can be effectively used in trading tasks of hedge funds, such as, in portfolio management both in traditional markets, including high-frequency trading, and in trading cryptocurrency assets, in modeling the market microstructure, market making, and other tasks [1], [2], [3], [4], [5], [6], [7], [8].

However, unlike a professional trader, an RL-based trading agent does not have prior knowledge of the subject area, let alone a basic experience or common sense. In the baseline formulation of the problem, the agent cannot decompose the trading universe into risk premiums. Moreover, sufficient time must be devoted to carefully adjusting the environment, defining the observation space, the reward function, and other hyperparameters to achieve acceptable results, particularly regarding solution stability.

Focusing on the outlined problems, an algorithm for optimizing the proximal policy is proposed and trading strategies (i.e., PnL structures) are proposed to optimize the choice of

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strategy in hedge fund management and asset trading with a predetermined base. Thus, the agent's controlling signal is the trading strategy. It should be understood that a trader's interaction with the environment is sequential and current decisions made by an investor can have a delayed medium- or long-term effect on the PnL structure. In contrast to several other approaches, reinforcement learning algorithms can accommodate such effects and produce an optimal trading strategy over the entire non-stationary trajectory.

The training of intelligent agents is an acute scientific and technical task of machine learning and has a wide range of applications, let alone the modern problems of financial science. This study focuses on the practical applicability of the proposed models and demonstrates the effectiveness of our approach through extensive backtesting and validation of historical market data, demonstrating superior performance compared to traditional methods.

The aim of this study is to develop and study an algorithm for integrating reinforcement learning (RL), particularly the proximal policy optimization (PPO) algorithm, into the process of selecting a hedge fund management strategy. The scientific novelty of this study lies in the development and validation of a scalable strategy-based framework for financial instrument trading using modern deep reinforcement learning algorithms. The theoretical significance of this study lies in presenting insights into the performance of machine learning based algorithms in trading, improving the understanding of PnL structure decomposition and methods for tuning the PnL structure. Using RL, the algorithm proposed in this study learns to adaptively adjusts strategy allocation to maximize total returns while respecting risk constraints.

The considered trading agents can be effectively used to solve the tasks of trading a financial instrument. In summary, this study offers new insights into the use of RL for optimizing strategy selection in hedge fund management and highlights the potential of AI-based approaches in finance.

Efficiency is assessed by the key performance indicators of the investment, particularly, the Sharpe ratio, CVaR, and CDaR. The effectiveness of the proposed algorithm was validated through extensive backtesting and verification of historical market data, demonstrating superior performance compared to traditional methods. The trading strategy formed by the proposed hedge fund trading agents has better performance indicators than alternative approaches, in particular, the basic strategies, that is, momentum, mean reversion, and buy-and-hold (B&H) strategy.

This study does not aim to propose a universal method for forecasting returns or developing highly optimized trading strategy (whether measured by the Sharpe ratio or other portfolio performance metrics) applicable to all financial instruments. Our goal is to develop an advanced approach for trading fundamental strategies underlying the algorithm strategies that are described and reproducible through the application of Reinforcement Learning (RL). We demonstrate that the proposed method is effective and outline the processes required to implement such an algorithm within a trading system.

II. LITERATURE REVIEW

Financial markets are experiencing a widespread increase in the use of automated trading methods such as algorithmic trading. The areas of algorithmic trading applications are expanding and their technology is improving. Recently, algorithmic trading methods have increasingly been adopted by hedge funds. The results confirmed the effectiveness of such solutions [9]. Therefore, it is necessary to further improve both the algorithmic trading technology itself and the methods and models of its application to hedge funds.

It should be noted that a large number of scientific studies in this direction are currently being conducted and published [10], [11], [12], [13], [14], [15]. These studies are aimed at developing a more effective trading strategy based on machine learning methods and algorithms, forecasting methods, and artificial intelligence. With the benefits of artificial intelligence, investors can use deep learning models to predict and evaluate the stock and forex markets. It should be noted that deep reinforcement learning algorithms are widely used in algorithmic trading. Improving the decomposition of the PnL structure (Profit and Loss), as well as methods for adjusting the PnL structure in machine-learning algorithms, is also relevant. The efficiency of algorithms is usually assessed using key investment performance indicators, particularly the Sharpe ratio (SR).Another fundamental issue emerging with machine-learning models, and reinforcement learning algorithms in particular, is the robustness of the underlying models and the interpretability of the results.

One of the popular areas of research in the field of algorithmic trading is the development of new deep reinforcement learning models for trading implementation. For example, the work [5] proposes a new deep reinforcement learning model for implementing stock trading. The research [16] proposes a trading algorithm employing TD3, which is a continuous action space Deep Reinforcement Learning (DRL), to allow the trader to buy/sell a dynamic number of shares.

The work in [12] presented the development of a stock investment robot using a Proximal Policy Optimization (PPO) reinforcement learning algorithm and the performance of the robot. In this case, two robots were used: the first robot used only stock data, a single stock, as input, and the second robot used stock data and index together with US interest rate data as input data. The stock investment performance of the two robots were comparatively analyzed using test data. Almost all developments in new deep reinforcement learning models for trading implementation require further improvement or larger-scale testing.

Paper [17] proposes the Momentum Investment Twin Delay Deep Deterministic (MITD3) policy gradient algorithm based on TD3. At the core of MITD3, a Momentum Investing critic (MI-critic) computes the Q-value according to the state-action pair and advantage historical performance of the actor, and gives a higher Q-value to encourage past winners and a lower value to penalize past losers. The work evaluates the advantage historical performance of the actor. The authors claim that proposed MITD3 outperforms state-of-the-art DRL approaches and generates intuitively explainable stock trading decisions. However, the proposed solution only takes into account profit indicators and does not take into account risks.

Along with various hybrid machine learning models, ensemble models have also been proposed. For example, proposed an ensemble trading strategy [18] using three actor critic based algorithms: Proximal Policy Optimization (PPO), Advantage Actor Critic (A2C), and Deep Deterministic Policy Gradient (DDPG). This strategy outperforms the three individual algorithms and two baselines in terms of the risk-adjusted return measured by the Sharpe ratio. Ensemble learning algorithms are distinguished by their complex technical and algorithmic implementation.

There are works proposed high-frequency trading models have been proposed [6] and show good results; however, these models are very complex in implementation. Algorithms that use a multi-agent reinforcement learning system to optimize financial trading strategies based on TimesNet [19] are innovative. This paper proposes a new approach called a multi-agent dual-deep Q-network within a multi-agent reinforcement learning framework by innovatively using two different agents. This proposal features a complex decision-making system that reduces the reliability.

Some studies have proposed various adaptive stock trading strategies using deep reinforcement learning methods. For example, in the work [14] are presented two trading strategies with reinforcement learning methods which are proposed as GDQN (Gated Deep Q-learning trading strategy) and GDPG (Gated Deterministic Policy Gradient trading strategy). A novel reward function is designed with a risk-adjusted ratio for the proposed GDQN and GDPG trading strategies to ensure stable returns, even in a volatile stock market.

One of the goals of using algorithmic trading is to eliminate traders' emotions in choosing strategies and making decisions. Therefore, to develop these systems, methods based on fundamental and technical analyses are mainly used, which are technically implemented using reinforcement learning models. This approach does not consider the skills of professional traders. Some recent studies imitate the decision-making models and trading philosophy of professional traders in stock trading [20]. However, this development requires further improvement before it can be proposed for practical applications.

It should also be noted that review works have appeared on the main areas of research and development of methods and models of algorithmic trading [3].

Considering the potential of previously explored AI-based approaches in finance, this study proposes a novel RL-based approach to optimize strategy selection in hedge fund management. At the same time, management considers not only the profit indicator but also risk, which is one of the main components of decision-making. This proposal differs from previous works in its relative versatility and provides a specific set of technologies and guidance for hedge fund managers, researchers, and practitioners who want to automate and operate their ML-based algorithmic trading developments.

III. METHODOLOGY

A. MATHEMATICAL MODEL

We formulate the strategy selection problem as an MDP, which is generally presented as a tuple $< \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R} >$, in where $\mathcal{S}$ is a finite set of states, $\mathcal{A}$ is a finite set of actions, $\mathcal{T}$ is a transition probability function and $\mathcal{R}$ is a reward function. The state space comprises market conditions, and the action space consists of strategy allocations. In this case, the states represent the set of trading strategies $S = \{S_1, S_2, \ldots, S_n\}$, each characterized by its respective parameters and historical performance metrics [21], [22].

Our study employs a partially observable Markov decision-making process (POMDP), defined as the 7-tuple:

$$< \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, \mathcal{R}, \mathcal{Z}, \gamma > . \tag{1}$$

$\mathcal{S}$ is a set of states that satisfying the Markov property, $\mathcal{A}$ is a set of actions, $\mathcal{O}$ is a set of observations, $\mathcal{P}$ is a state transition model, $\mathcal{R}$ is a reward model satisfying the Markov property, $\mathcal{Z}$ is an observation model, formally, $\mathcal{Z}^a_{s'o} = \mathbb{P}\left[O_{t+1} s_t = s', a_t = a\right]$ and $\gamma = [0, 1]$ the discount factor.

From the point of view of the subject area, both the return from time t 1 to t and more complex statistics, such as the Sharpe ratio, CVaR, and CDaR, as well as their combinations, can be used as a reward.

The objective is to develop a policy that maximizes cumulative returns while respecting risk constraints.

Let us consider the main components of the model in relation to our research.

B. KEY CONCEPTS

The environment is a dynamic system in which agent learning occurs. In this study, the system is the financial market, specifically the US stock market, as part of FAANG.

In a practical sense, the environment is an interface with certain methods, including, but not limited to performing a step in the environment, that is, applying some action setting the environment to the initial state and representing the state of the environment in a visual form (to render). Taking a step implies that the environment should return to the agent's observation, reward, whether the current state is terminal, and auxiliary information for diagnostics.

The agent is a controller that interacts with the environment. From a practical point of view, the main function of the agent is to select the optimal action (or set of actions) based on the available observation o_t. In this study, the agent is defined as the trading agent itself, that is, the agent that selects strategies.

Action is the control signal of the agent $a_t \in \mathbb{A}$ at time t. Action is the only method for the agent to have an impact on the environment. In the given task, the agent's action is to trade one of the basic strategies, the momentum or mean-reversion strategy.

The agent state s_t^a is the agent's internal representation of the current state of the environment s_t^e to make a decision a^t written as a function

$$s_t^a = f(h_t). \tag{2}$$

where $h_t = (o_1, a_1, r_1, \ldots, o_t, a_t, r_t)$ is the sequence of observations, actions, and rewards received by an agent in the process of interaction with the environment from the initial state up to including time t.

In a fully observable environment $s_t^a = s_t^e = o_t$, that is the agent has access to the state of the environment in its entirety in the form of observation. This study assumes that complete observability is unrealistic for the financial market. Additionally, the state of the environment may contain incorrect information.

In the context of finance, both price data and its derivatives, along with alternative indicators can be used to shape the state of the agent, such as fundamental data (dividends, multiples, etc.), aggregates of company sentiment (news publications, etc.), derivative data such as option lines as well as volumes and spreads.

A model of state transitions can be identified as

$$\mathcal{P}_{ss'}^a = \mathbb{P}\left[s_{t+1} = s' \mid s_t = s, a_t = a\right]. \tag{3}$$

while the rewards model can be formulated as

$$\mathcal{R}_s^a = \mathbb{E}[r_{t+1} \mid s_t = s, a_t = a]. \tag{4}$$

Usually, reinforcement learning tasks are reduced to MDP or POMDP.

State s_t is said to satisfy the Markov property if and only if (the future is independent of the past and determined by the present)

$$\mathbb{P}[s_{t+1} \mid s_t, s_{t-1}, \ldots, s_1] = \mathbb{P}[s_{t+1} \mid s_t]. \tag{5}$$

The policy is a function $\pi : \mathbb{S} \rightarrow \mathbb{A}$ which describes the agent's behavior. Two types of strategies are recognized: deterministic and stochastic. Thus, the first strategy can be written as $a = \pi(s)$.

Stochastic strategies represent a probability distribution on a set of actions under the current state s:

$$a = \pi(a \mid s) = \mathbb{P}[a_t = a \mid s_t = s]. \tag{6}$$

The state-value function allows the agent to estimate the usefulness of the agent in the current state:

$$V^\pi = \mathbb{E}_\pi\left[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \ldots \mid s_t = s\right]. \tag{7}$$

The action-value function allows an agent to estimate the usefulness of taking a certain action in some state and following the existing policy in the future:

$$Q^\pi = \mathbb{E}_\pi\left[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \ldots \mid s_t = s, a_t = a\right]. \tag{8}$$

Transactional costs are those accompanying order execution, such as brokerage commissions (including interest in margin transactions and uncovered positions) and slippage (bid-ask spreads).

Assessment of transaction costs is an extremely important task of backtesting the developed model as the investment advantage of a strategy that does not consider transaction costs found on historical data can be offset by commissions or slippage when testing the model in the market.

Slippage is the deviation of the actual execution price of an order to buy or sell a security from the price at which the investor is expected to open the transaction.

In this study, transaction costs and slippage were estimated as a constant percentage deviation from the target, that is, the observed, closing price on a given trading bar.

Risk is the potential of adverse consequences associated with owning a particular instrument or asset. When analyzing strategies in options trading, three types of risks can be defined: gamma risk, which refers to the risk of rapid deterioration of conditions; vega risk, which pertains to the risk of future overestimation or underestimation; and jump or proxy 'skew' risk, which involves the risk of sudden adverse changes in conditions. These risks provide a proximate assessment of the strategy's Profit and Loss (PnL) structure, even if options are not directly involved. This is the framework within which this study conceived, the selection of momentum, and mean-reversion strategy sets justified by the diverse quasi-gamma risk of these strategies.

C. TRADING STRATEGY ESSENCE

A great variety of trading strategies and underpinning models exist to address market inefficiencies or participate in market movements, such as arbitrage, carry trading, earnings announcements strategies, anomaly detection trading, market timing, momentum, mean-reversion, pairs trading, seasonality strategies, sentiment trading, spread trading, trend-following strategies, volatility trading (including volatility selling), and volume effects trading. Several possible classifications of trading strategies according to various criteria can be presented [23], [24].

The most frequently used strategies are momentum and mean-reversion strategies. The scientific community, traders and investors have long been aware of the momentum effect and have found that this effect occurs in a wide variety of markets and time frames [25], [26], [27]. Using an extensive dataset of different asset classes, Moskovitz demonstrate that the returns over the past 12 months predict the next month's returns; a trading strategy that buys assets if their returns over the past 12 months are positive and sells those assets otherwise generates significant risk-adjusted returns [28]. However, it should be understood that the particular configuration of a momentum strategy may vary from study to study, from one practical implementation to another. A specific set of parameters is predominantly determined by a particular trading universe, time frame, and so on.

Momentum strategies aim to profit from persistent price trends, yet occasionally these trends collapse. The mean - reversion strategy aims to profit from the tendency of asset prices to return to the trend, that is, to the mean [29].

Equity return momentum in the short-run and reversal in the long-run are two prominent financial market anomalies. If one optimally incorporates both return momentum and reversal into a trading strategy, one would expect to outperform the strategies based only on return momentum or reversal, and even the buy-and-hold strategy.

D. PERFORMANCE METRICS

The advantage of an investment cannot be assessed in terms of individual performance indicators, it is critical to assess performance from different angles, for example, using the following metrics:

1) Cumulative Return (%) is a measure of the total change in value of an investment or portfolio over a specific period, expressed as a percentage of the initial value: 1.5 meaning 150% of returns or 2.5x yield. It is calculated by taking the current value of the investment, subtracting the initial value, and dividing by the initial value. This metric helps investors assess the overall performance of an investment strategy.
2) Market Value ($) is the total value of an asset or portfolio at a given point in time, typically calculated by multiplying the current price of the asset by the number of shares or units outstanding. In the context of long/short exposure, it reflects the total investment value in both long positions (buying assets expecting their prices to rise) and short positions (selling borrowed assets expecting their prices to fall). Long/Short Exposure refers to an investment strategy that involves taking long positions in assets expected to appreciate and short positions in assets expected to depreciate. Later in the text, these and other metrics will be used in tables and charts.
3) Drawdown (%) refers to the decline in the value of an investment or portfolio from its peak to its lowest point during a specific period, expressed as a percentage: -0.5 meaning 50% of drawdown. It indicates the extent of loss an investor would have experienced and is often used to measure risk. A larger drawdown suggests higher risk, as it shows a more significant loss from the highest value achieved.
4) Conditional Drawdown at Risk (CDaR) is presented by Chekhlov and Uryasev [30]. CDaR has the advantage of assessing all drawdowns, not only the drawdown with the maximum value:

$$CDaR_\alpha(X) = DaR_\alpha(X) + \frac{1}{\alpha * T} * \sum_{i=0}^{T} \max\Bigg[\max_{t\in(0,T)}\Bigg(\prod_{i=0}^{t}(1+x_i)\Bigg) - \prod_{i=0}^{j}(1+x_i) - DaR_\alpha(X), 0\Bigg] \in \mathbb{R}. \tag{9}$$

where

$$DaR_\alpha(X) = \max\Bigg(DD(X,j) \in \mathbb{R} : \mathcal{F}_{DD}DD(X,j) < (1-\alpha)\Bigg) \in \mathbb{R}.$$

5) Conditional Value at Risk (CVaR) is a measure of risk that quantifies the amount of tail risk of an asset or strategy [31]:

$$CVaR_\alpha(X) = VaR_\alpha(X) + \frac{1}{\alpha * T}\sum_{t=1}^{T}\max(-X_t - VaR_\alpha, 0) \in \mathbb{R}. \tag{10}$$

where

$$VaR_\alpha(X) = -inf_{t\in(0,T)}\,(X_t \in \mathbb{R} : F_x(X_t) > \alpha) \in \mathbb{R}.$$

where α is given probability level, X_t - asset returns.
6) The reward function of the agent will be defined as the Sharpe ratio over a certain period of time. The quality of investment decisions made by a trader directly depends on the degree of relevance of the selected indicators.

IV. ALGORITHM DEVELOPMENT

Our algorithm leverages the PPO algorithm, which is a state-of-the-art RL technique, to learn an optimal strategy selection policy. At each time step, the algorithm observes the current state of the market, selects actions corresponding to strategy allocations, executes trade, and receives feedback in the form of rewards.

The policy is updated iteratively based on the observed rewards, with the objective of maximizing cumulative returns over time. Risk management considerations were integrated into the reward function to ensure prudent portfolio management.

A. KEY POINTS

At the top level of abstraction, trading a financial instrument using a trading strategy can be considered a partially observable Markov decision process (POMDP). In general, this process can be represented as shown (Fig. 1).

The agent observing the state of the system makes a decision on the choice of a trading strategy, for example, one of the two conventional profit and loss (PnL) structures depending on gamma risk decomposition that can be obtained on an asset, that is, to buy/sell when the asset rises or falls (momentum strategy) or to buy/sell when the asset falls/rises

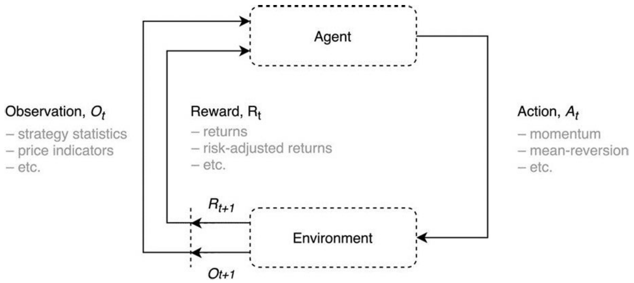

FIGURE 1. **Model of a financial instrument trading process.**

(mean-reversion strategy). By following the trading rules of the strategy, the agent is rewarded with returns, for example, in the most general sense. The process then repeats itself.

The most advanced and consistent trading of a financial instrument requires the consideration of the factors and axes of risk that drive the returns for which an investor is rewarded. This challenge is admittedly greater because researchers and investors do not agree on a unique set of factors for any given asset class. For example, in the general case, the asset's expected excess return (over cash) can be represented as follows:

*Asset expected excess return (over cash) = asset growth factor beta * growth premium + asset inflation beta * inflation premium + asset illiquidity beta * illiquidity premium + …*

It should be noted that the majority of models, especially those based on machine learning and reinforcement learning, offered by the scientific community and industry experts lack this insight. The algorithm does not know and is unable to grasp where asset returns come from, what risk premiums are driving them [32], [33], and eventually just pursues returns or other targets. Hence, the algorithm can converge to a solution with overloaded factors that carry a larger return or Sharpe ratio at the expense of certain risk premia. For example, at the cost of the tail risk premium. Such a result or solution of the algorithm is perilous and can break the structure and risk metrics of the portfolio, that is, the returns on which the trader or client focuses. It is especially dangerous when tail risk is realized, a factor that can be overloaded in the portfolio. Nevertheless, this framework can be used to make sense of and structure the tasks at hand.

The advantage of our model lies not in the application of Reinforcement Learning to trading tasks, but in the management of a portfolio of strategies and the expansion of such algorithms beyond merely determining which asset to purchase. It encompasses selecting a strategy or deciding on which manager to invest. Our innovation consists in trading not individual assets, but rather a set of strategies. This approach has not been previously considered and enables the operation and scaling of large trading blocks. We compare strategy trading with the strategies themselves to demonstrate that our model outperforms any individual strategy. There are no other comparable approaches for our algorithm; trading individual assets using Reinforcement Learning constitutes an entirely different challenge.

Further in the paper, separate stages of this algorithm will be considered in more detail. Given the specifics of the problem, this study used model-free algorithms to evaluate both the utility function and strategy. In this study, the PPO algorithm trains the trading agent and identifies the optimal trading strategy for the short-term investment horizon. We consider the most prominent reinforcement-learning algorithm in more detail.

B. PPO ALGORITHM

One of the disadvantages of reinforcement learning as an approach is that the learning sample generated by the agent while interacting with the environment directly depends on the current policy such that the distributions of observations and rewards are constantly changing as the agent learns, which generates instability in the learning process. In addition, reinforcement learning methods are very sensitive to hyperparameters such as policy initialization.

The PPO algorithm, a proximal policy optimization algorithm (Fig. 3), was introduced by the OpenAI team in 2017 to address the outlined problems [34].

The policy gradient methods to which this algorithm belongs have the problem of sample inefficiency, because the experience accumulated by the agent becomes irrelevant after the current strategy (the gradient descent step) is updated.

The key advantage of PPO is that it ensures that the new agent's strategy is not significantly different from the previous one by limiting the ratio $r_t(\theta) = \frac{\pi_\theta(a_t|S_t)}{\pi_{\theta_k}(a_t|S_t)}$ — of the updated strategy to the old strategy, such that it is in the range of 1.0. The algorithm does not require calculation of the Kullback–Leibler distance, as proposed in a similar algorithm, TRPO, which is the main method under consideration.

The target optimization function $\mathcal{L}_{\theta_k}^{CLIP+VF+S}(\theta)$ is a linear combination of the modified target function of the gradient policy methods $r_t(\theta) * A_t^{\pi_k}$, shifting the strategy towards actions with a greater advantage (a higher reward than expected for the average state), the target state value function $V_\theta(s_t)$, used to calculate advantages, and entropy $\mathcal{S}_{\theta_k}(\theta)$ to provide the agent with a study of the environment.

Consequently, the algorithm is relatively simple to implement, sampling efficiency is achieved, and the hyperparameters are relatively easy to tune.

V. VALIDATION AND PERFORMANCE ANALYSIS

We conducted extensive backtesting and validation using historical market data across diverse asset classes and market conditions. A comparative analysis against conventional methods demonstrates the superior performance and adaptability of our RL-based approach in selecting optimal trading strategies.

A. BACKTESTING METHOD

Backtesting is an important stage in the development of an effective trading strategy, preceding trading on a demo account and real trading.

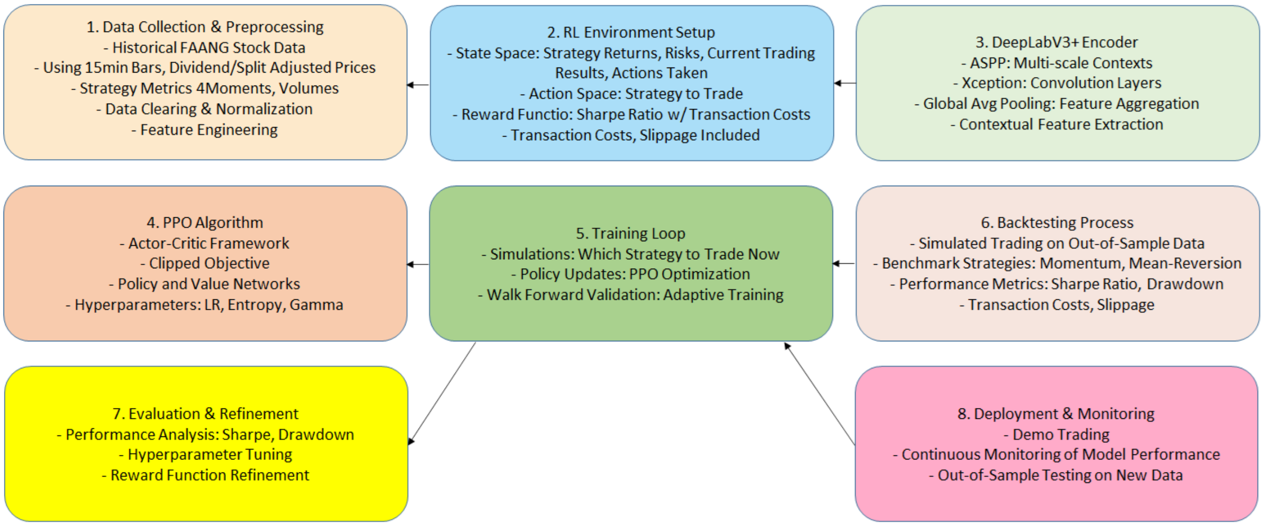

FIGURE 2. **Algorithm for selecting the optimal trading strategy based on reinforcement learning.**

Algorithm PPO

Given: initial policy parameters $\boldsymbol{\theta}_0$, coefficients $\boldsymbol{c}_1, \boldsymbol{c}_2$

for $k = 0,1,2,\ldots$ **execute**

Collect multiple partial trajectories $\mathcal{D}_k$ with policy $\boldsymbol{\pi}_k = \boldsymbol{\pi}(\boldsymbol{\theta}_k)$

Estimate advantages $\widehat{A}_t^{\pi k}$

Calculate new policy parameters

$$\boldsymbol{\theta}_{k+1} = \arg\max_{A} \mathcal{L}_{\theta_k}^{CLIP+VF+S}(\boldsymbol{\theta})$$

by optimization with SGD (Adam) with K epochs and mini-batch of size M of surrogate function $\mathcal{L}$, where

$$\mathcal{L}_{\theta_k}^{CLIP+VF+S}(\boldsymbol{\theta}) = \mathcal{L}_{\theta_k}^{CLIP} - c_1\mathcal{L}_{\theta_k}^{VF} + c_2 S_{\theta_k}(\boldsymbol{\theta}),$$

$$\mathcal{L}_{\theta_k}^{CLIP}(\boldsymbol{\theta}) = \mathbb{E}_{\tau\sim\pi_k}\left[\sum_{t=0}^{T}\left[min\left(r_t(\boldsymbol{\theta})\widehat{A}_t^{\pi k}, clip\left(r_t(\boldsymbol{\theta}), 1-\epsilon, 1+\epsilon\right)\widehat{A}_t^{\pi k}\right)\right]\right],$$

$$\mathcal{L}_{\theta_k}^{VF}(\boldsymbol{\theta}) = \mathbb{E}_{\tau\sim\pi_k}\left[\sum_{t=0}^{T}\left[\left(V_\theta(s_t) - V_t^{target}\right)^2\right]\right], r_t(\boldsymbol{\theta}) = \pi_\theta\left(a_t|s_t\right)/\pi_{\theta_k}\left(a_t|s_t\right)$$

end

FIGURE 3. **Algorithm of proximal policy optimization.**

The backtesting process covers the simulation of a sequence of trades, i.e. trades on historical or simulated data. Historical data and simulations enable the calculation of strategy statistics to assess the effectiveness of the strategy as well as to evaluate possible scenarios, both positive (what returns an investor can expect) and negative (possible drawdowns, losses and risks) (Fig. 4).

The backtesting process is used to identify drawbacks and limitations, to optimize and fine-tune the investment strategy, and build confidence in the underlying decision-making model before launching the strategy in production. On the other hand, it should be understood that superior model performance on historical or simulated data cannot guarantee comparable results on a real trading account. Several issues are closely related to the backtesting process, including the poor quality of the historical data source and backtesting platform. In addition, critical mistakes can be made in the backtesting process, including during the data preparation stage. Separately, we should mention systematic survivorship bias, incorrect sample handling (data snooping bias and data leakage), and the problem of overfitting machine learning models.

A certain toolkit exists to address the described nuances, including both general methods, such as regularization,

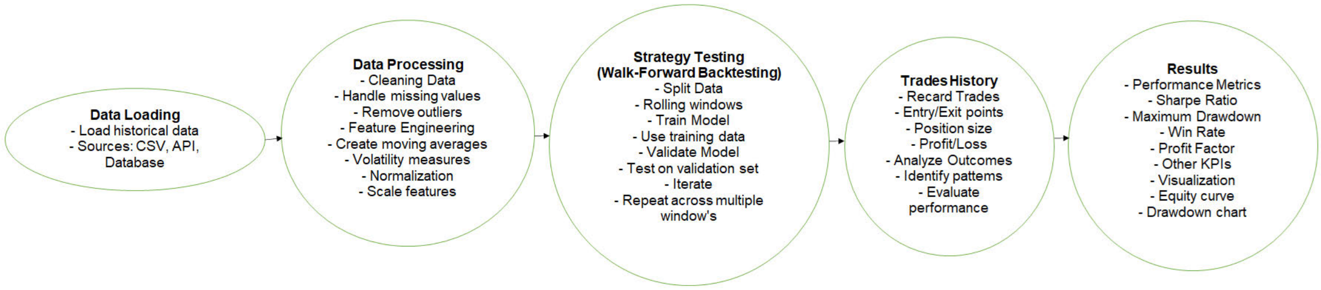

FIGURE 4. Strategy performance assessment backtesting pipeline.

model complexity limitation and cross-validation, and special methods such as White's Reality Check (Bootstrap Reality Check) [35]. However from a practical point of view, the most reliable test for ''reality'' is testing a model by trading on a demo or real account.

B. ASSUMPTIONS

The indicators observed by the agent at time t do not contain any future information. In particular, normalization, averaging and calculation of other aggregating statistics are performed in the sliding window so that the agent has no knowledge of the future even in an implicit form. Otherwise, the decision- making model formulated by the trading agent is biased.

Further, we assume sufficient liquidity of traded assets; that is, that at any time t the agent can sell the selected security at a price close to the market price. The main argument in favor of the assumption noted in this study can be the significant capitalization of traded companies.

In addition, the absence of delay, slippage, and influence of the agent's actions on the market when opening and closing the positions of the portfolio is assumed, that is at the closing of the trading bar one can instantly place orders, which will be immediately executed at the specified price. Brokerage commissions, borrow fees, and slippage during order execution were evaluated separately and included in the PnL reports.

To account for corporate events (payment of dividends, use of splits, mergers and acquisitions, realization of poisoned pills, etc.) affecting the quotes, adjusted prices are used to calculate the returns, commissions, etc.

When dealing with the price data of instruments as well as with their derivatives, for example, fundamental multipliers, it is extremely important to consider the influence of corporate events. For example, if the issuer decides to make a 1:6 split on the ex-date the value of the security decreases six times, which is apparently not a market movement.

C. MODEL CONFIGURATION

To solve the problem of trading a financial instrument, particularly the optimal allocation of funds to one of the strategies, the action space is defined by one of the strategies $S_t : S_t \in (Momenum, Mean - Reversion)$.

The observation space is represented by a tensor $o_t \in \mathbb{R}^{M+T}$ containing the original dataset described above, where M is the number of indicators and T is the historical observation window, that is, the lookback window.

In other words, at the time of trading strategy selection for a particular instrument, the agent has access to information about the risk-metrics of a given asset as well as the strategy in different windows, the first four moments of the distributions of returns in different windows, volumes, etc. up to a certain point in time t in the past. Presumably, this model configuration allows the agent to assess the attractiveness of both the instrument and use of a particular strategy.

Determining the reward is the most significant and challenging step in designing intelligent agents. This study uses the Sharpe ratio, taking into account transaction costs over a certain window as the reward.

Next, we consider the configuration of the PPO algorithm presented above, implemented using the TensorFlow library. The algorithm was based on the DeepLabV3+ encoder architecture adapted for the task (Fig. 4), which has shown excellent results in semantic segmentation (separation of objects by their structure before recognition) [36].

A distinctive feature of the DeepLabV3+ architecture is the utilization of atrous spatial pyramid pooling, which allows the encoding of contextual information at different scales by applying extended atrous convolution layers with different parameters, ''fields of view''. In addition, the Xception subgraph is an advanced approach in image classification, batch normalization to regularize and improve convergence, global average pooling to aggregate selected contexts, and a number of other techniques.

We utilize the encoder component of the algorithm in such a way that the global polling data is fed into the PPO model at the end, facilitating decision-making regarding which strategy is currently the most advantageous.

The motivation for choosing this architecture is to extract the investment quality of the asset as well as the strategies themselves in different economic contexts. The test period covers the 2020-01-01 – 2021-08-01 range in 15 min frequency, characterized by the global stock market crash that started on February 20, 2020, in the wake of the coronavirus pandemic and subsequent quantitative easing

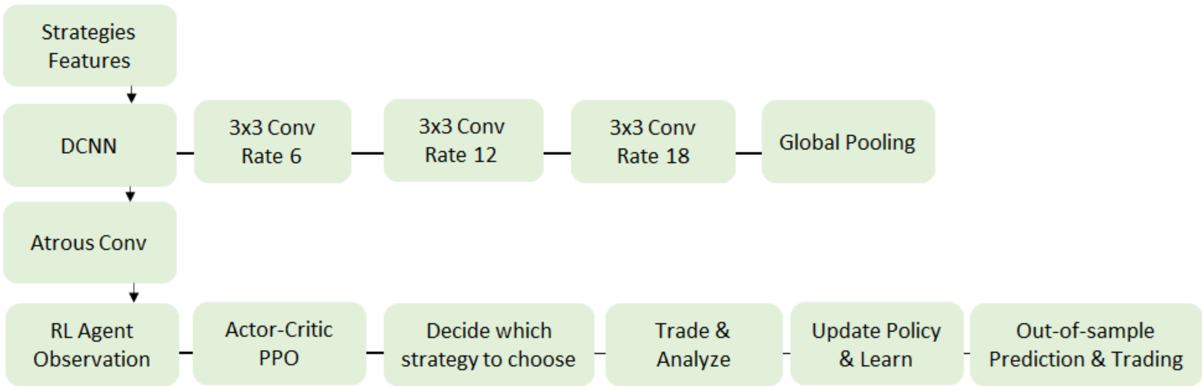

FIGURE 5. DeepLabV3+ network architecture. This paper uses only the adapted subgraph of the network responsible for encoding.

TABLE 1. Limitations of the research.

Limitation	Definition
Data	Data quality and data processing procedures
Universe	FAANG Stocks: FB, AMZN, AAPL, NFLX, GOOG
Backtesting	T-0 order execution, liquidity, no influence on the market
Benchmarks	Momentum, Mean-Reversion, Buy-and-Hold

(QE) and interest rate cuts, and the US stock market recovery and growth period.

The frequency of strategy selection is limited to a trading bar of 15 minutes, allowing the agent to make timely short-term investment decisions. Validation of the models is carried out using the rolling out-of-sample method, which is the closest to real trading conditions. This method assumes that once a period, e.g., once a day, the weights of the model are completely erased, the learning process starts from scratch, on the data available at that time. Then, as backtesting progresses, the agent makes decisions based on the newly observed information.

Transaction costs include trade commissions, the cost of securing a short position, and slippage, determined by some coefficient (percentage of price deviation).

The benchmarks are the two basic strategies momentum and mean-reversion as well as the buy-and-hold strategy.

The basic momentum and mean-reversion strategies are constructed as holding or selling periods depending on the $z - score$ of the asset returns over a certain horizon. For example, for the momentum strategy the trading range is limited to an interval of $[0.1, 1.5]$ for buying and $[-0.1, -1.5]$ for selling. For the mean-reversion strategy: $[-1.5, -0.1]$ and $[0.1, 1.5]$ respectively.

Notably the parameters of the basic strategies were not been optimized. A single heuristic set of parameters was used for all assets to reflect the essence of strategy performance but not to optimize their returns and other parameters. The task of the agent is to learn how to derive an investment advantage, even from non-optimized trading strategies.

VI. PRACTICAL IMPLICATIONS

A. DATA

The processes for uploading and processing raw financial data are subject to rigorous testing and validation as the results of the study are predominantly determined by the correctness of the observations; the main provider of the data is Bloomberg. The initial dataset comprises price data, volumes, returns density function statistics, volatility estimates, risk metrics, moving autoregression ratios, autocorrelation function estimates, and derived indicators for each stock.

Trading universe. To test the developed models, the historical data of five US stocks, namely the components of the FAANG acronym, were used. Validation of the selected assets implies a certain physical process, which may be partially or fully irrelevant to other assets, such as futures contracts or options.

Backtesting assumptions. Backtesting of historical data implies $T - 0$ delay in order execution, liquidity of traded instruments, and lack of trading agents' influence on the market (in particular, the order book). A separate task is to evaluate brokerage commissions, transaction costs associated with opening short positions, and slippage when placing orders. Dividends were not reinvested during backtesting.

Computational power. This study as well as reinforcement learning algorithms in general is limited by the significant computational complexity of the models used and the cost of appropriate resources for their computation (including for business), and the fact that model training can cost millions of dollars is often overlooked in studies of such agents [37].

Alternative approaches and benchmarks. The securities portfolio management methods chosen for efficiency comparison may be inferior to more comprehensive management approaches such as models based on advanced statistical approaches and genetic algorithms. Nevertheless, the alternative approaches under consideration provided an estimate of the relative effectiveness of the algorithms.

Real trading. To validate the developed models in full, it is necessary to evaluate the effectiveness of trading agents

FIGURE 6. Discounted rewards distributions at different points during the training.

on demo and real accounts, to compare the results with the results of the backtest for the corresponding period, which is beyond the scope of this study.

The following methods and technologies are used to solve the formulated tasks:

- The main source of price and alternative data in this paper is Bloomberg.
- The high-level general-purpose programming language Python as well as the libraries NumPy, Pandas (data processing), TensorFlow (algorithms development), etc. were used to implement the environment and trading agents.
- Backtest calculations on the full dataset were performed on a remote server (16 vCPU, 64 GB RAM).

B. RESULTS

The estimated backtesting time of the models was one month. It has been shown that the developed trading agents can be effectively used to handle the problem of basic strategy trading in the framework context, namely momentum and mean-reversion, in terms of the Sharpe Ratio, CVaR, CDaR metrics. In addition, the ultimate strategies formulated by the agent have a higher performance than the baseline strategies themselves, as well as the buy-and-hold strategy (Table 1).

The 95% confidence intervals for the RL strategies for the presented metrics for the different strategy runs are shown. The figures indicate the average performance of the strategies in the different runs.

In this respect, the best results in terms of the Sharpe Ratio in the roll out-of-sample approach have been achieved on assets with higher performance on the underlying strategies and on the assets themselves, that is, on inherently more stable assets with higher rates of return per unit of risk, such as the AAPL instrument. Nevertheless, the performance of the underlying asset is not a required condition for constructing an optimal trading strategy. An agent can explore the context of strategies and predict their viability at the time of trading

TABLE 2. Performance risk metrics for the RL strategy, underpinning strategies of the algorithm, namely Momentum, Mean-Reversion, and the B&H strategy for different assets.

	FB		
Strategy	**Sharpe ratio**	**CVaR**	**CDaR**
RL	**1.76 ± 0.21**	**0.53 ± 0.25**	**0.17 ± 0.18**
Buy-and-hold	1.04	0.65	0.51
Momentum 1	0.39	0.52	0.46
Momentum 2	0.03	0.54	0.51
Mean-Reversion 1	-2.34	**0.33**	0.38
Mean-Reversion 2	-0.48	0.41	0.30
	AMZN		
Strategy	**Sharpe ratio**	**CVaR**	**CDaR**
RL	**1.75 ± 0.27**	**0.48 ± 0.21**	**0.20 ± 0.12**
Buy-and-hold	1.26	0.76	0.32
Momentum 1	-0.82	0.46	0.44
Momentum 2	0.36	0.53	0.44
Mean-Reversion 1	0.05	**0.27**	**0.17**
Mean-Reversion 2	0.48	0.36	0.30
	AAPL		
Strategy	**Sharpe ratio**	**CVaR**	**CDaR**
RL	**2.90 ± 0.18**	**0.51 ± 0.25**	**0.18 ± 0.07**
Buy-and-hold	1.37	0.82	0.35
Momentum 1	0.07	0.49	0.30
Momentum 2	1.71	0.54	0.33
Mean-Reversion 1	-2.13	**0.30**	0.37
Mean-Reversion 2	-1.10	0.45	0.45
	NFLX		
Strategy	**Sharpe ratio**	**CVaR**	**CDaR**
RL	**2.38 ± 0.13**	**0.55 ± 0.18**	**0.22 ± 0.09**
Buy-and-hold	0.66	0.87	0.39
Momentum 1	-1.25	0.57	0.55
Momentum 2	-0.78	0.62	0.64
Mean-Reversion 1	-0.64	**0.34**	0.26
Mean-Reversion 2	0.95	0.41	**0.19**
	GOOG		
Strategy	**Sharpe ratio**	**CVaR**	**CDaR**
RL	**1.90 ± 0.26**	**0.46 ± 0.16**	**0.13 ± 0.04**
Buy-and-hold	1.70	0.56	0.29
Momentum 1	-1.48	0.47	0.63
Momentum 2	0.77	0.42	0.19
Mean-Reversion 1	-0.83	**0.26**	0.17
Mean-Reversion 2	-1.27	0.42	0.40

FIGURE 7. Discounted rewards distributions at different points during the training.

despite the fact that the total return of a strategy may even be negative, e.g., FB or GOOG.

In addition, it must be understood that the proposed algorithms are not the best solution and can be improved and calibrated as part of the optimization of the problem in terms of the bias-variance trade-off.

In terms of tail risk, namely the CVaR and CDaR metrics, in one of the realizations (upper bounds of the confidence

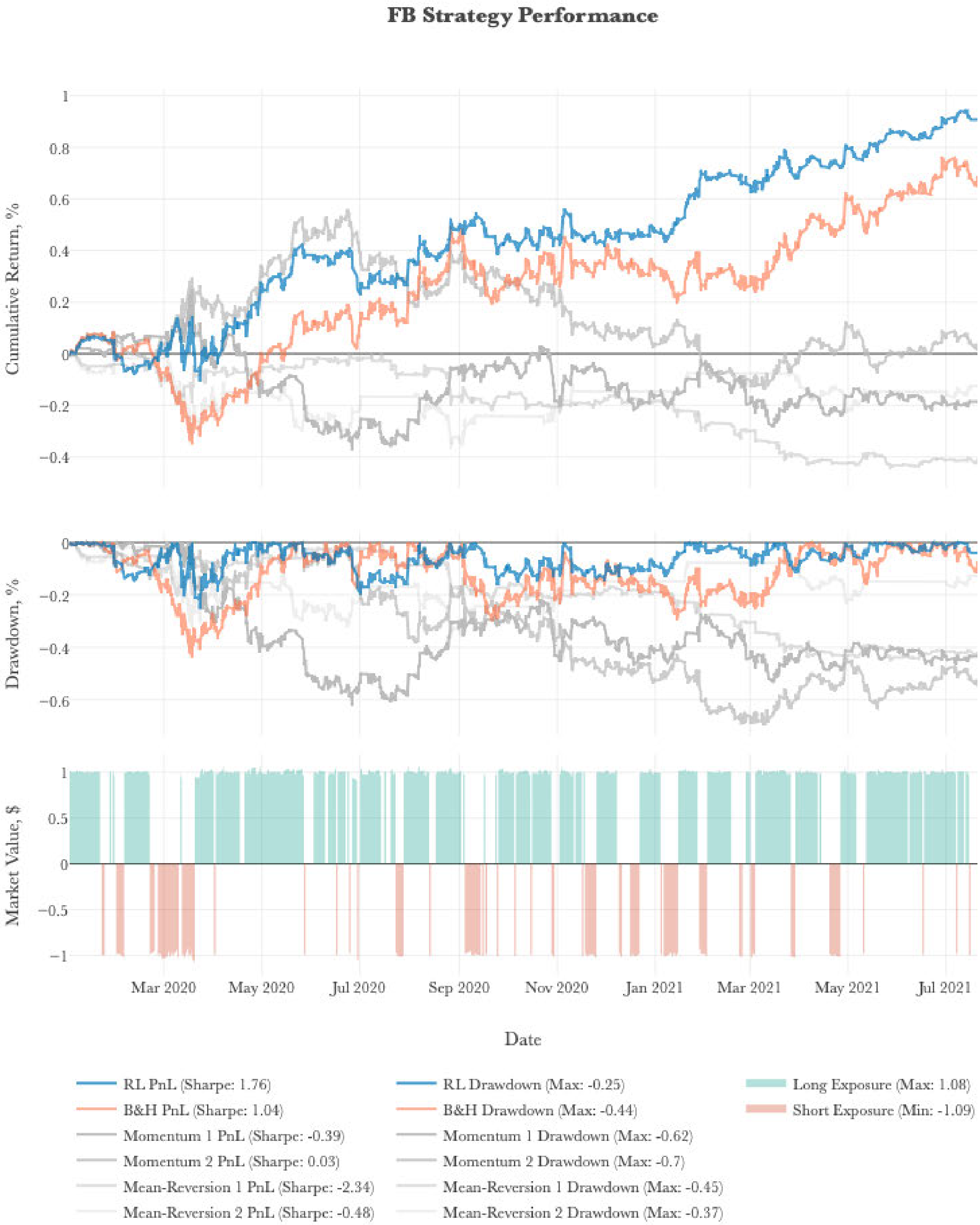

FIGURE 8. Report on the performance of the RL strategy on FB shares.

interval) the algorithm may be inferior to alternative strategies, yet the overall return per unit risk of RL-based strategies is substantially higher.

Another point to note is the effectiveness of the PPO algorithm based on the DeepLabV3+ network for the task at hand. A neural network can deduce the financial context of strategy exploitation and select the optimal trading strategy for the nearest trading horizon. Furthermore, the algorithm can infer the impact of transaction costs on the cumulative performance of the resulting strategy and limit frequent

FIGURE 9. Report on the performance of the RL strategy on AMZN shares.

strategy switching despite the high frequency of data, namely 15 min.

Subsequently, we report the performance of the algorithms for each asset individually. The cumulative returns including transaction costs, drawdowns and net exposures (market values) of the RL strategy, the underlying momentum, mean-reversion strategies with different z-score windows, and the buy-and-hold strategy are presented.

C. LEARNING PROCESS

The learning process of the algorithms is considered equally important. To address this problem, a several tools exist

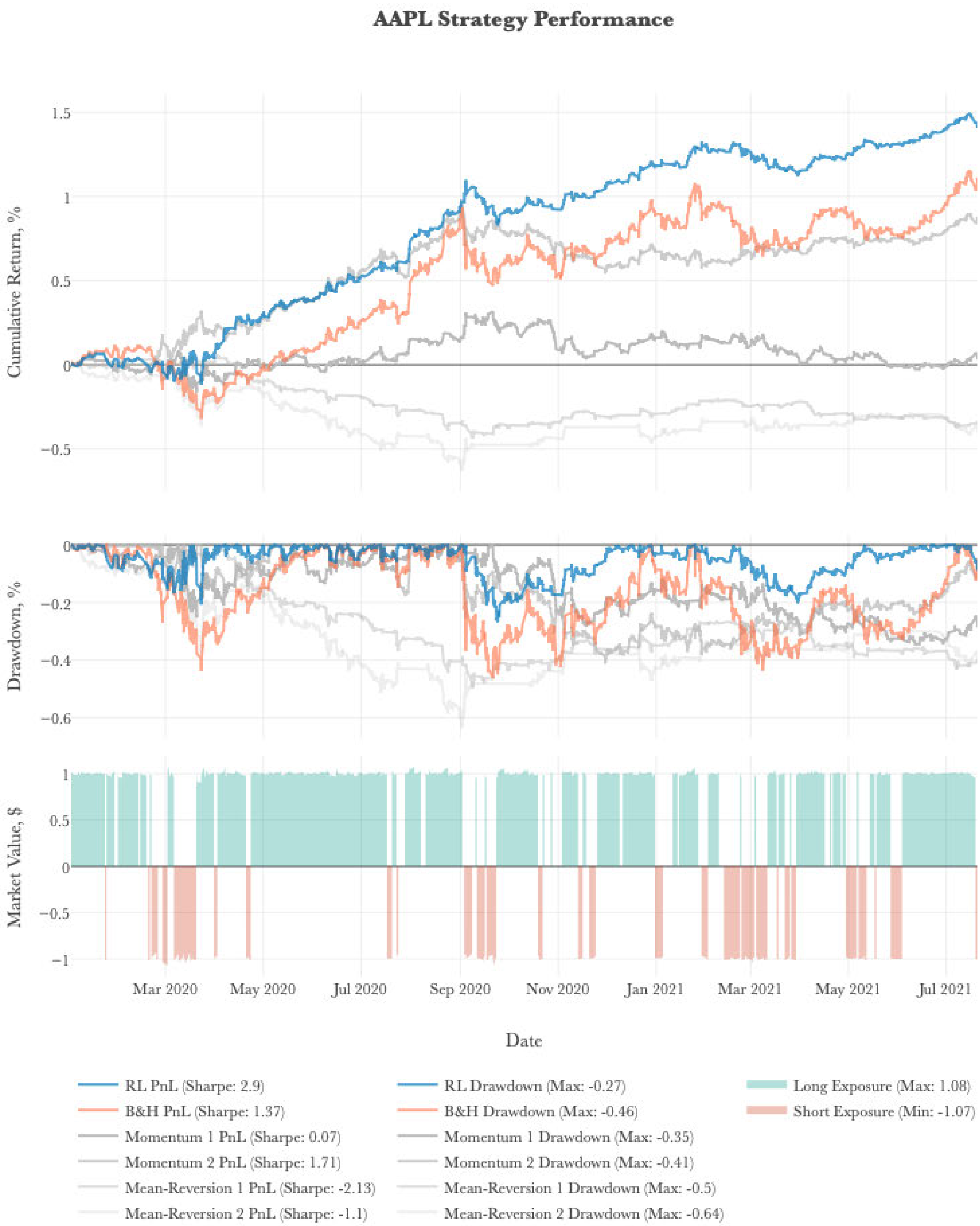

FIGURE 10. Report on the performance of the RL strategy on AAPL shares.

to evaluate the performance of the model in real time. For example, the TensorFlow library provides toolkits for tracking scalar metrics such as loss and accuracy, etc. which assist in deepening the understanding of model performance. At the same time, such utilities allow you to abort the training process to save computational resources if no quality gain is observed for a significant time, that is the neural network is unable to disentangle the data structure and retrieve the mapping from the observation space to the action space in an adequate time frame.

FIGURE 11. Report on the performance of the RL strategy on NFLX shares.

Discounted reward distributions help visualize the learning process of models, showing how rewards evolve over time (Fig.6 and Fig. 7) the axes represent the reward values, illustrating how the distribution of rewards shifts as learning progresses. In Fig. 7, the x-axis represents the number of learning steps, providing insight into how rewards accumulate as the model trains. This visualization indicates that the models are indeed learning, as the patterns in the reward distributions reveal improvements in performance over time.

The provided graph (Fig. 7) shows an example of the learning process of the model at various points. It can be

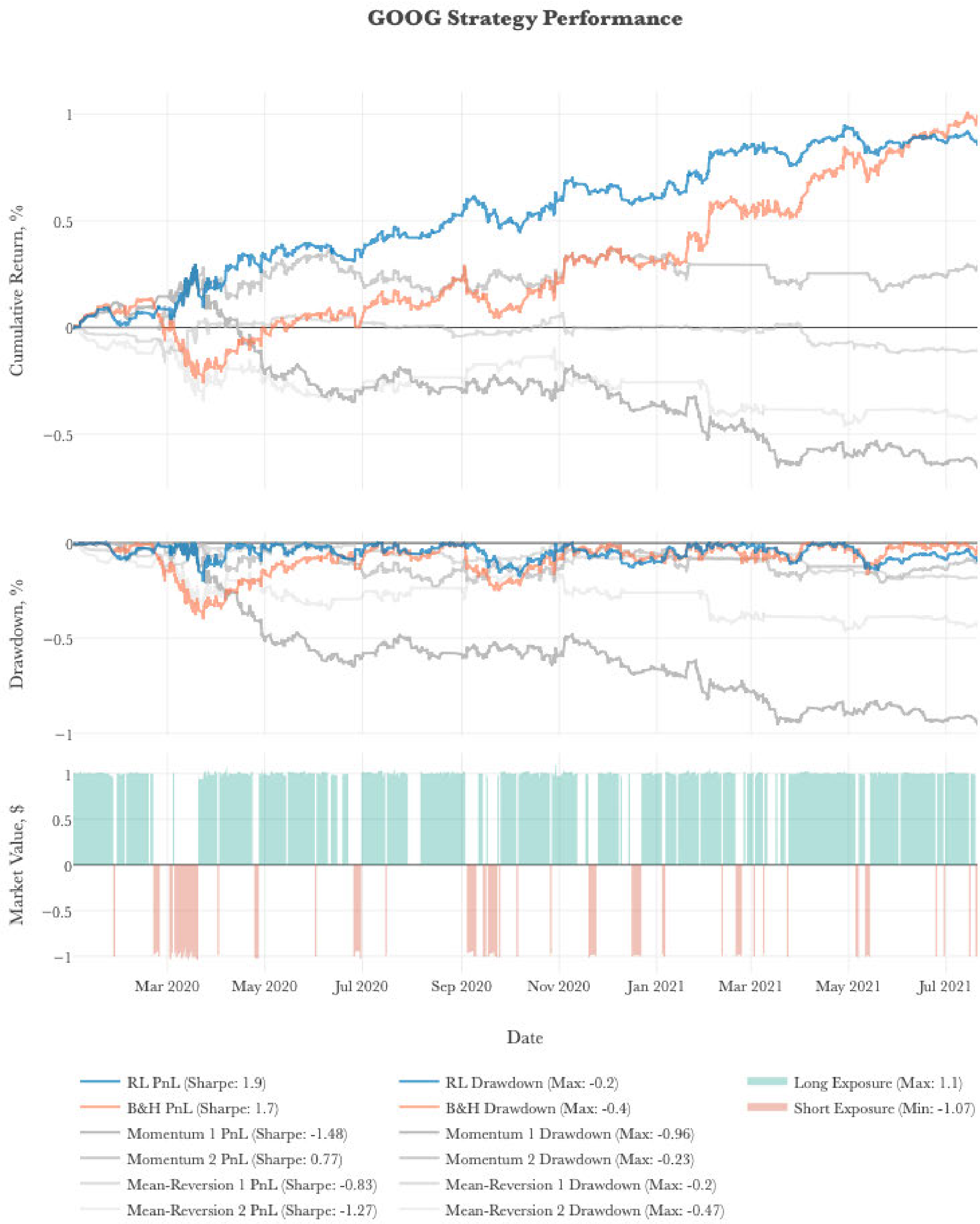

FIGURE 12. **Report on the performance of the RL strategy on GOOG shares.**

seen that initially the distribution of discounted rewards is multimodal and located predominantly on the left-hand plane (the area of the least rewards of the agent). As learning progresses, the PPO algorithm shifts the reward to the positive side, with the tails of the distributions shrinking and the multimodality disappears.

The presented charts (Fig. 8 - Fig. 12) demonstrate the core learning principle of gradient-based policy algorithms, namely the shift of the reward distribution in the positive direction. Interestingly, it is notable that the reward increases drastically after 50000 iterations of the learning process, and then increases steadily thereafter.

Currently, the fundamental problems of machine learning and reinforcement learning preclude the complete delegation of complex intelligence tasks faced by humans, such as investment asset trading, to state-of-the-art deep learning algorithms with reinforcement. In addition, the algorithm does not understand the structure of asset risk premiums and is incapable of managing the PnL structure on its own.

Unlike a professional trader, the algorithm has no a priori knowledge of the subject area, much less basic experience and common sense, is unable to understand at the expense of which risk premium is achieved, and which risk-axis drives the reward. Despite this, by limiting the action space and allowing the agent to follow one of the basic strategies and risk premiums, the algorithm is trained within a robust framework specified by the human, who is more fully aware of the subject area.

The performance reports of the RL strategy for each asset individually are presented in (Fig. 8 - Fig 12). The cumulative returns including transaction costs, drawdowns and net exposures (market values) of the RL strategy, the underlying momentum, mean-reversion strategies with different z score windows, and the buy-and-hold strategy are presented.

VII. CONCLUSION AND FUTURE DIRECTIONS

In conclusion, we present a pioneering RL-based approach to optimize strategy selection in hedge fund management. Our research underscores the potential of AI-driven approaches in finance and opens avenues for further exploration, including the incorporation of additional factors and integration of RL with other advanced techniques.

In this study we evaluated the performance of state-of-the-art deep reinforcement learning algorithms in framework-based asset trading tasks. It is shown that the strategies produced by the developed trading agents are efficient and have statistically significantly higher performance than alternative trading strategies such as momentum, mean-reversion, and buy-and- hold. Nevertheless, it is important to note that this approach has certain drawbacks, including those related to the specifics of the task domain. First, the training of trading agents requires a considerable amount of time, computing and other resources. In addition, it should be noted that the model parameters need to be adjusted carefully owing to their intrinsic sensitivity to hyperparameter changes.

In addition, to achieve the main objective of this, the essence of asset ownership and strategy-based trading, methods of performance evaluation are analyzed; modern algorithms of deep reinforcement learning and applicability to financial tasks are studied and the trading environment is developed and tested. We compare strategy trading with the strategies themselves to demonstrate that our model outperforms any individual strategy. It would also be interesting to evaluate the applicability of other Reinforcement Learning algorithms in this context.

One of the fundamental problems of machine learning models and reinforcement learning algorithms is the trustworthiness of the underlying models and the interpretability of the results. To this end, it is suggested that further work entails a comprehensive analysis of the results in terms of specific market situations, both positive and negative, a substantive study of the principles of sellers, and hypotheses for improving the algorithms based on the analysis.

Additionally, ''black box'' interpretation methods can also be used to address interpretation issues. For example, it has been proposed to estimate individual conditional expectation (ICE), plot partial dependence plots (PDP), and determine SHAP values of features using the SHapley Additive exPlanations method [10]. Another area of focus is code optimization and model adapting for GPU computation.

Further, it is feasible to extend or partially replace the trading set of securities, including other types of instruments and attributes, to build the agent's observation space, as well as the mechanisms of reward function formation.

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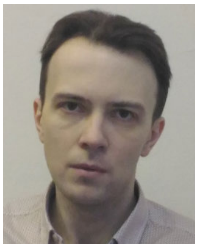

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